

ApplyFlow AI - Software Requirements Specification (SRS)

1. Project Overview

Project Name

ApplyFlow AI

Project Type

AI-Powered Job Application Tracking & Legacy System Modernization Platform

Goal

Transform a legacy C++ job application tracker into a modern AI-powered web platform using ASP.NET Core, PostgreSQL, Next.js, AI Agents, Jira Automation, and CI/CD pipelines.

The system will help users:

- Track job applications
 - Manage interviews and offers
 - Generate insights using AI
 - Automate project management workflows
 - Demonstrate legacy modernization capabilities
-

2. Technology Stack

Backend

- ASP.NET Core Web API
- Entity Framework Core
- PostgreSQL
- JWT Authentication
- BCrypt Password Hashing

Frontend

- Next.js
- TypeScript
- Tailwind CSS

- Shadcn UI

AI Layer

- Claude
- OpenAI APIs
- AI Agents

Project Management

- Jira Scrum
- Sprint Planning
- Story Tracking

DevOps

- GitHub
 - GitHub Actions
 - Docker
 - CI/CD
-

3. User Roles

Job Seeker

Can:

- Register
- Login
- Manage applications
- Track interviews
- View analytics

Admin

Can:

- Manage users
 - View platform statistics
 - Manage system settings
-

4. Functional Requirements

Module 1: Authentication

User Registration

User can:

- Create account
- Enter email
- Enter password
- Create profile

API:

POST /api/users/register

User Login

User can:

- Login using email and password

API:

POST /api/users/login

JWT Authentication

System must:

- Generate JWT tokens
 - Protect private endpoints
 - Validate user identity
-

Current User

User can fetch profile

API:

GET /api/users/me

Module 2: Application Management

Create Application

User can:

- Add company name
- Add job title
- Add application date
- Add notes

API:

POST /api/applications

View Applications

API:

GET /api/applications

View Single Application

API:

GET /api/applications/{id}

Update Application

API:

PUT /api/applications/{id}

Delete Application

API:

DELETE /api/applications/{id}

Module 3: Status Tracking

Application statuses:

- Applied
- Screening
- Interview
- Offer
- Rejected
- Withdrawn

User can update status anytime.

Module 4: Dashboard

Display:

- Total Applications
 - Active Applications
 - Rejections
 - Offers
 - Interview Count
-

Module 5: Search & Filters

Filter by:

- Company
- Position
- Status
- Date Range

Search by:

- Company Name
 - Position Title
-

Module 6: AI Features

Resume Analysis

User uploads resume.

AI provides:

- Strengths
 - Weaknesses
 - ATS Score
 - Improvement Suggestions
-

Job Match Analysis

AI compares:

Resume vs Job Description

Returns:

- Match Score
 - Missing Skills
 - Recommendations
-

Application Insights

AI analyzes:

- Success Rate
 - Common Rejection Patterns
 - Interview Trends
-

Module 7: Legacy Modernization Showcase

Purpose:

Demonstrate transformation from legacy system to modern architecture.

Stages:

1. Legacy C++ System
 2. Requirements Extraction
 3. User Story Generation
 4. Modern Architecture Design
 5. ASP.NET Core Migration
 6. Database Migration
 7. AI Integration
-

Module 8: Jira Integration

Scrum Workflow

Epic → User Stories → Tasks → Sprint

Sprint Tracking

Support:

- Backlog
 - Active Sprint
 - Sprint Review
 - Burndown Tracking
-

Module 9: AI Project Automation

Future Feature

Workflow:

User Story → AI Agent → Code Generation → Pull Request Draft → Review

Example:

"As a user I want login functionality"

AI automatically generates:

- DTOs

- Controllers
 - Services
 - Unit Tests
-

5. Database Entities

Users

- Id
 - Email
 - PasswordHash
 - FirstName
 - LastName
 - CreatedAt
-

Applications

- Id
 - UserId
 - CompanyName
 - PositionTitle
 - ApplicationDate
 - Status
 - Notes
 - CreatedAt
-

6. MVP Scope

Phase 1

Must Complete:

- Authentication
 - JWT
 - Application CRUD
 - Status Tracking
 - Dashboard Statistics
 - PostgreSQL Integration
 - Swagger Documentation
-

7. Phase 2

- Resume Analysis
 - Job Match Analysis
 - AI Insights
 - Email Notifications
 - Docker Deployment
-

8. Phase 3

- Jira Automation
 - AI Agent Workflow
 - Story-to-Code Generation
 - Automated Pull Requests
 - CI/CD Automation
-

Success Criteria

A user should be able to:

1. Register
2. Login
3. Create applications
4. Track application status
5. View dashboard analytics
6. Receive AI insights
7. Demonstrate a complete legacy-to-modern migration workflow